

The empirical degree for OEIS A201351, proved from an exact model

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14 September 2026

Abstract

OEIS A201351 carries the empirical claim that its terms agree with a polynomial of degree 63. It is true, and it is decidable rather than empirical. The claim names no coefficients, so there is no formula to check; but a sequence agrees with a polynomial of degree at most d on a range exactly when its $(d + 1)$ -st finite difference vanishes there, and the degree is exactly d exactly when the d -th difference does not eventually vanish. Both are linear recurrences, and the entry's count is given by an exact model that provably satisfies a monic linear recurrence of order $S = 2631$, so both are settled by a finite exact computation.

2020 Mathematics Subject Classification. 05A15, 05A16, 11B37.

1 The conjecture

OEIS A201351 is “Number of $n \times 6$ 0..1 arrays with rows and columns lexicographically nondecreasing and every element equal to at least one horizontal or vertical neighbor.”. It has offset 1 and begins

5, 36, 273, 2732, 29540, 310036, 3030673, 27104947, ...

The entry states, as a conjecture contributed by its author and never marked settled:

Empirical: $a(n)$ is a polynomial of degree 63 for $n > 22$

As of the “Last modified” line on the live entry (Oct 03 2025 09:05:14, revision 9) this is still recorded as empirical, and nothing on the entry records it as proved.

2 The count is a walk count

The entry's defining condition is local: it is a condition on the previous row together with, for each adjacent pair of columns, whether they still agree and which way their latest difference went, and, for each position, whether its obligation to equal a neighbour is still outstanding. Take as vertices the admissible configurations of such a window and put an edge from u to v when v may follow u , which the condition decides by inspection. An admissible array is then exactly a walk, so with M the adjacency matrix of that digraph on $S = 2631$ vertices and ι, τ the vectors recording which windows may begin and end an array, $a(n) = \iota^\top M^{g(n)} \tau$ for the linear function g the entry's shape supplies. In particular a is C -finite.

3 A degree is two recurrences

Lemma 1. *Let Δ be the forward difference. For integers N and $d \geq 0$, a agrees on $n \geq N$ with a polynomial of degree at most d if and only if $(\Delta^{d+1}a)(n) = 0$ for every $n \geq N$; and that polynomial has degree exactly d unless $(\Delta^d a)(n) = 0$ for all large n as well.*

Proof. Standard finite differences. Δ^{d+1} kills exactly the polynomials of degree at most d , and on a range where $\Delta^{d+1}a$ vanishes the values of a are determined by $d+1$ consecutive ones through the same identity a degree- d polynomial satisfies, so the interpolating polynomial through any $d+1$ of them reproduces all of them. Its leading coefficient is $(\Delta^d a)/d!$, which is the constant the second statement is about. $\square$

Written out, $\Delta^{d+1}a = 0$ is the monic linear recurrence with characteristic polynomial $(z-1)^{63+1}$, and $\Delta^d a$ constant is the one with $(z-1)^{63}$. Both are fixed recurrences, so each is decided by the annihilation test: M^S is an integer combination of $I, \dots, M^{S-1}$ by Cayley–Hamilton, the residuals therefore obey the monic recurrence given by the characteristic polynomial of M , and S consecutive zeros force every later one.

4 The theorem

Theorem 2. *$a(n)$ agrees with a polynomial in n of degree exactly 63 for every $n \geq 23$.*

Proof. The residual of $(z-1)^{63+1}$ was formed at each index in exact integer arithmetic from the model. It is nonzero at $n = 86$ and vanishes from $n = 86 + 1$ onwards, with the run of zeros continuing past $S + 63 + 1$, which by the annihilation bound settles every later index at once. So $\Delta^{63+1}a$ vanishes on $n \geq 86 - 63 = 23$, and Lemma 1 gives the polynomial there.

For the degree: the residual of $(z-1)^{63}$ has no run of $S + 63$ zeros anywhere in the computed range, so $\Delta^{63}a$ is not eventually zero and the degree is not smaller than 63. The entry claims it for $n > 22$, and the threshold above is exactly $22 + 63 + 1 = 86$, so the range proved is the range claimed and neither is slack. $\square$

5 Verification

First, the model reproduces all 19 terms the entry publishes, exactly, in integer arithmetic; this is what ties it to the entry, which is read in English and could be misread. Second, both residuals were computed term by term rather than through any closed form. Third, the derived bound S was tested on terms it had not been given: the model was evaluated beyond the S values that determine it and the annihilator reproduced them.

References

- [1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequence A201351.
- [2] R. P. Stanley, *Enumerative Combinatorics*, Volume 1, 2nd ed., Cambridge University Press, 2011.
- [3] R. L. Graham, D. E. Knuth and O. Patashnik, *Concrete Mathematics*, 2nd ed., Addison–Wesley, 1994. (Finite differences.)